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Amrita Vishwa Vidyapeetham, Coimbatore Campus

B.Tech Degree Examinations – August 2026

Fifth Semester

Computer Science and Engineering

23CSE304 Embedded Systems

Duration: Two hours Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Understand the fundamentals, functions, and architecture of embedded systems.
CO02	Develop applications making use of GPIOs and peripheral units using Embedded C.
CO03	Develop applications for the usage of Timers, Interrupts, and Debugging of embedded applications.
CO04	Design an embedded system using an ARM processor.

Answer all questions

PART A (Theory)[20 Marks]

1) A modern automobile uses electronic systems for cruise control, engine management, safety, and entertainment. Explain the role of embedded systems in these functions and describe the characteristics required for their effective operation.

[6] [CO01][BTL 2]

2) How does the ARM Cortex-M4 use the Link Register (LR) during subroutine calls?

[3] [CO01][BTL 2]

3) State one function of the SysTick timer in ARM Cortex-M4. Calculate the reload value to be loaded into the SysTick Reload Value Register (SYST_RVR) to generate a periodic interrupt every 1 ms, assuming a processor clock of 16 MHz with a /8 prescaler

[3] [CO03][BTL 2]

4)

a. Apply the concept of bit-banding in ARM Cortex-M4 to control a GPIO pin using STM32F401CCU6. The GPIOB base address is 0x40020400 and the GPIO output data register (ODR) offset is 0x14.
   i. Determine the address of the GPIOB_ODR register and calculate the bit-band alias address corresponding to bit 4 of the register.
   ii. Explain how bit-banding improves GPIO control in real-time applications.

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[4] [CO02][BTL 3]

b. You are working with a 32-bit control register in the STM32F401CCU6 microcontroller. The register controls multiple hardware features through individual bits, and several bits are already set.
   i. Analyze the bitwise operation required to set bit 4 while preserving the states of all other bits in the register
   ii. Compare this approach with the use of a direct assignment such as REG = 0x10;, in terms of its effect on the other bits of the register.

[4] [CO02][BTL 4]

PART B (Practical)[30 Marks]

5) Design and implement a **Smart Fan Speed and Status Indicator system** on an STM32F401CCU6 microcontroller. The hardware setup is as follows: a switch is connected to PA6 to act as the fan power control, a potentiometer is connected to ADC1 channel 0 to regulate fan speed, another switch is connected to PB10 for manual override, and LEDs are connected to PC13, PC14, and PB9 for system indication. The system must operate in the following manner:
   a. when the switch on PA6 is pressed, it simulates turning the fan ON and the LED on PC13 should glow, while releasing the switch turns the fan OFF and the LED goes OFF.
   b. The potentiometer reading determines the fan speed level, which should be indicated by the blinking frequency of PB9 with two bands only: slow blinking for low speed and fast blinking for high speed.
   c. The manual override switch on PB10, when pressed, should force PB9 to stay steadily ON regardless of the potentiometer input, and simultaneously PC14 should glow steadily to indicate that override mode is active. Releasing PB10 must resume potentiometer-based automatic fan speed control. Importantly, if the fan is OFF (PA6 not pressed), both PB9 and PC14 must remain OFF irrespective of potentiometer or override input.

Rubrics:

Register-Level Configuration	[5] [CO02][BTL 2]
Door Sensor Functionality (PA6 → PC13)	[5] [CO02][BTL 3]
Potentiometer-Based Speed Indication (PB9 Blink, two bands)	[5] [CO02][BTL 3]
Manual Override (PB10 → PB9 steady ON + PC14 indicator)	[5] [CO02][BTL 3]
ADC Value Demonstration & Functional Verification	[10] [CO02][BTL 4]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**9**	BTL 1	
CO02	**38**	BTL 2	**17**
CO03	**3**	BTL 3	**19**
CO04		BTL 4	**14**
CO05		BTL 5	
CO06		BTL 6	

Bloom's Taxonomy Levels (attached for reference)

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply

Level 4 - Analyze

Level 5 - Evaluate

Level 6- Create

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